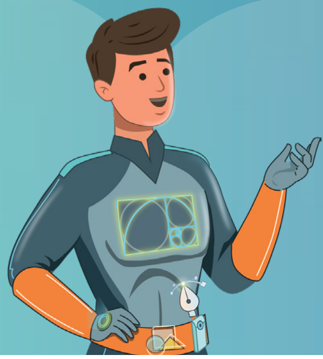
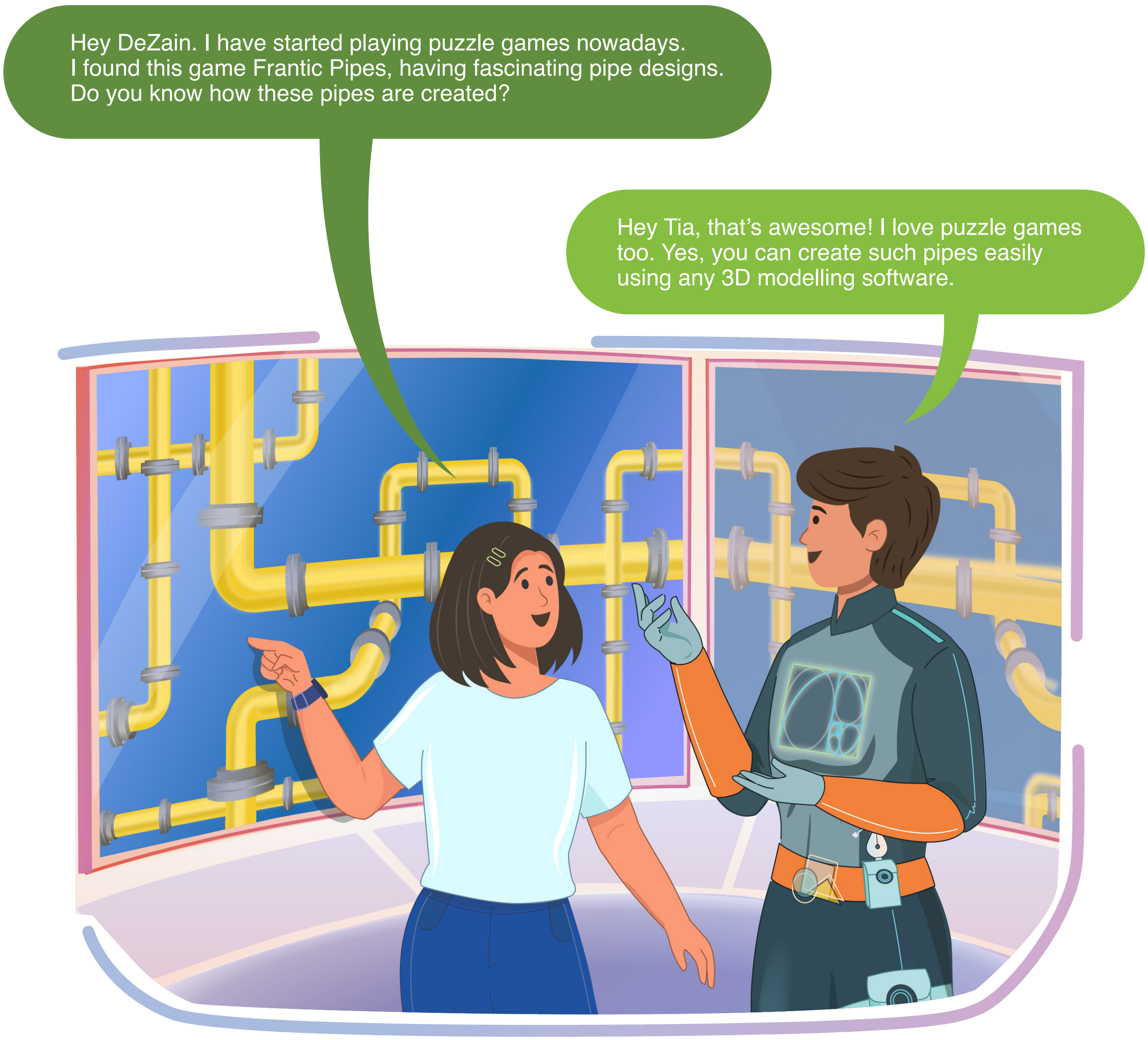
Objects Around Us - Part 2

10

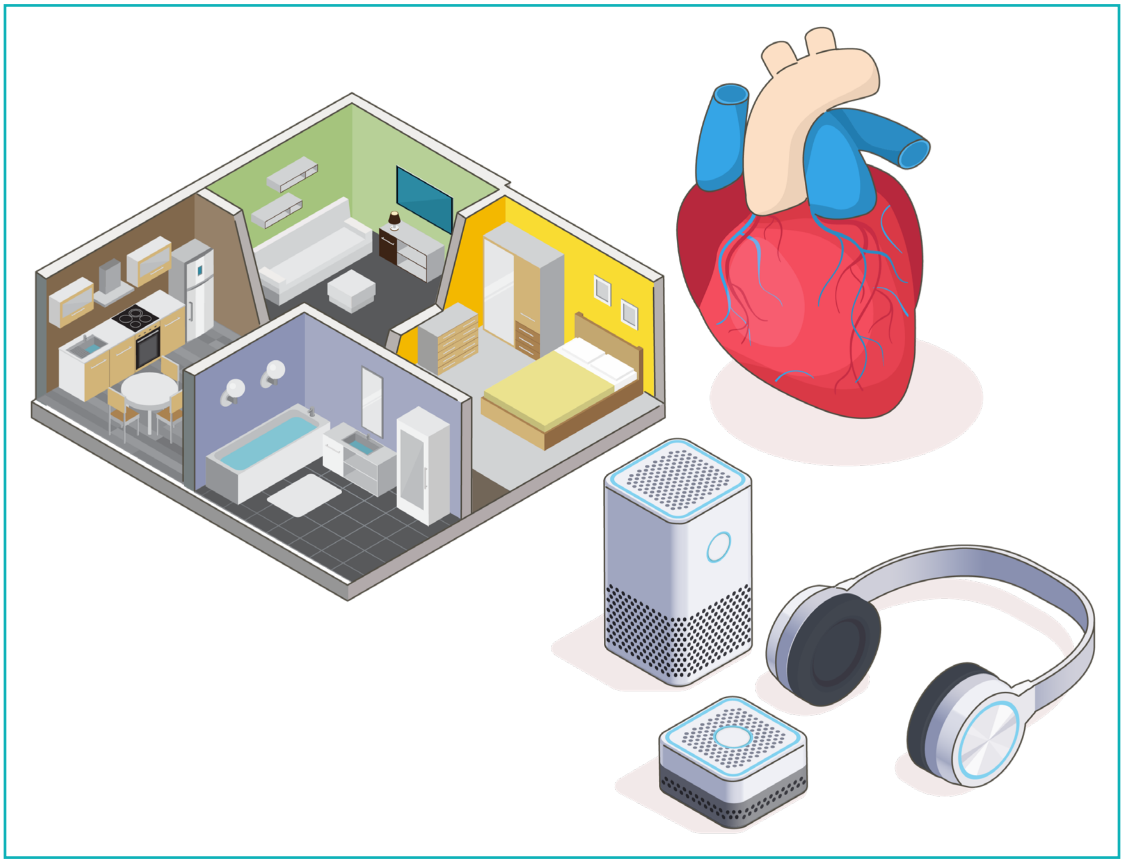




In Lesson 1, we studied the fundamentals of 3D modelling. It helps in developing our creativity and imagination. 3D modelling has given a chance to individuals to perceive their hobbies as career prospects. Let us understand more about how 3D modelling benefits students.

3D modelling helps us to understand complex concepts in physics, mathematics, chemistry, biology, and create science projects. It boosts creativity by allowing us to design and build our own models. We can utilise it to use our imaginations and create original concepts and creations. It encourages us to pay great attention to every detail in the models, which enhances attention to detail. 3D modelling enhances spatial thinking.

3D modelling prepares us for a variety of prospective careers, including architecture, engineering, product design, and animation. It provides us with hands-on exposure to a technology that is increasingly in demand in the job market today.

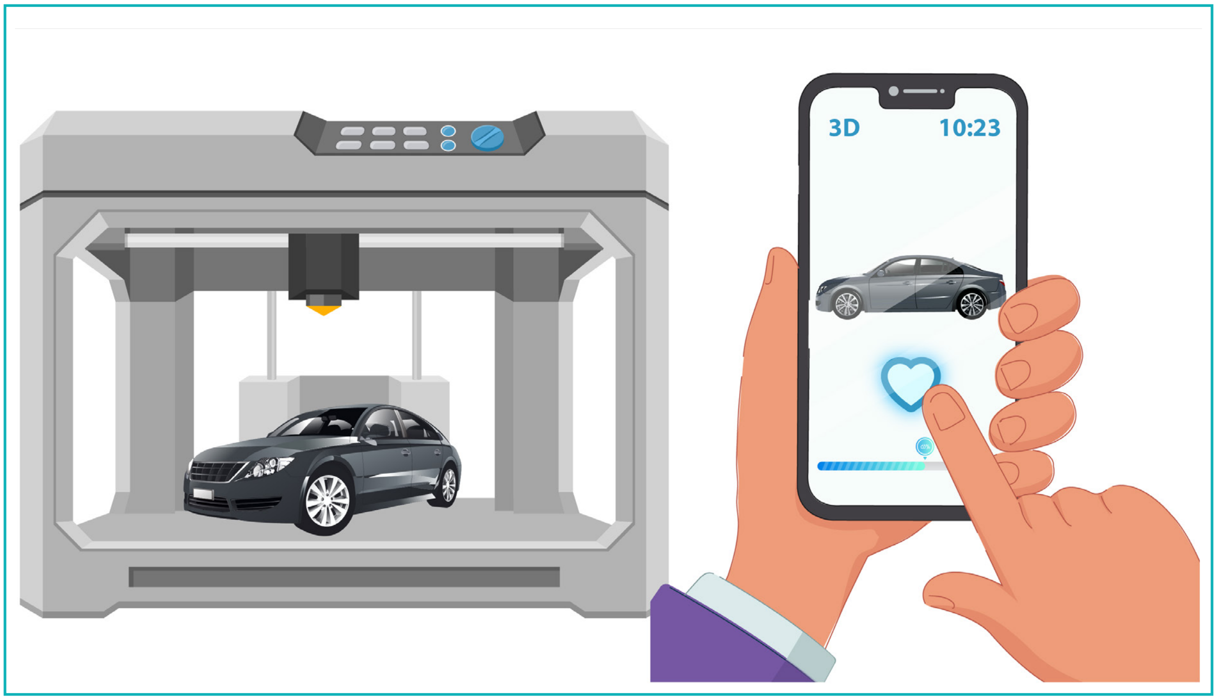


3D Printing

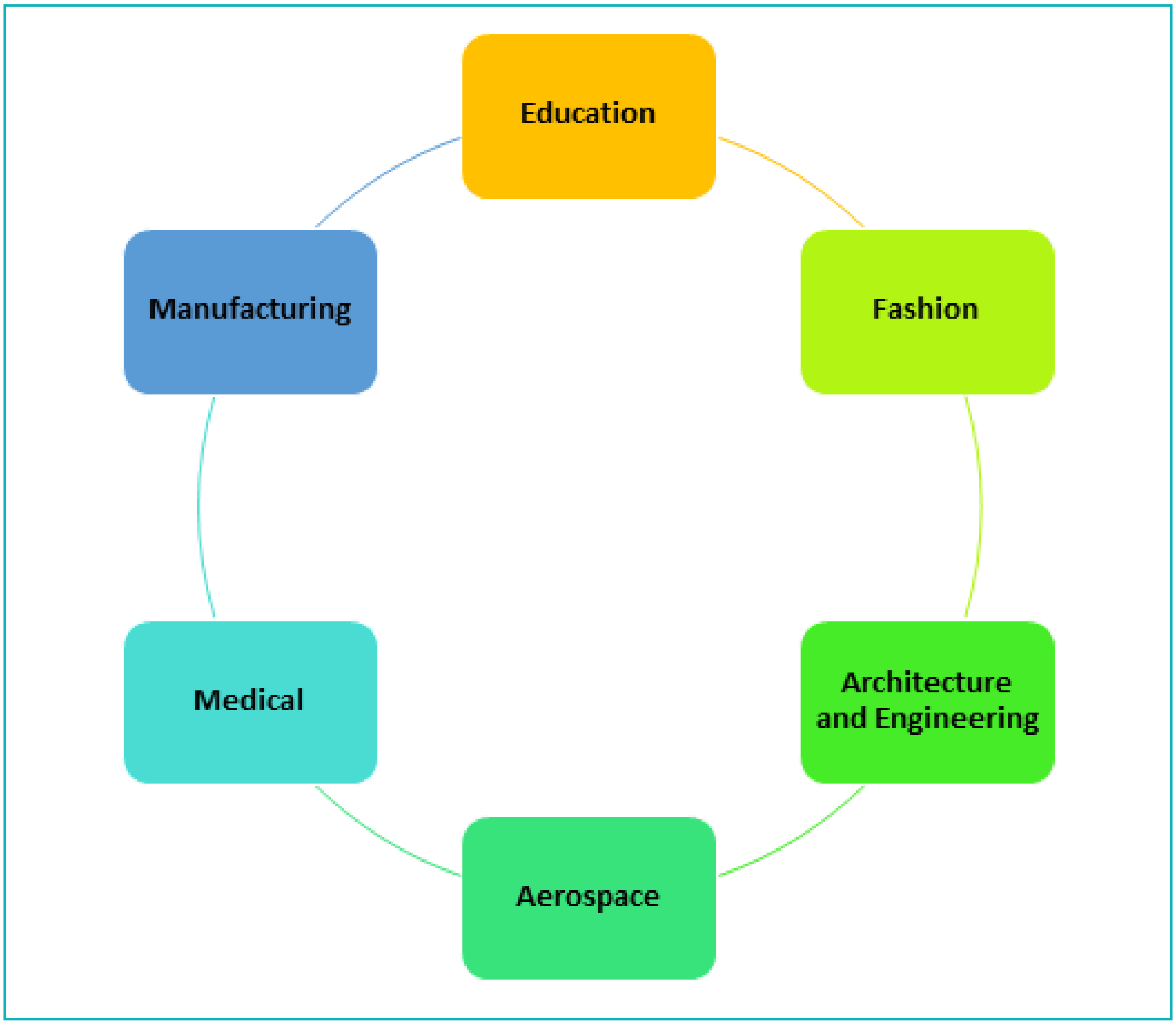
3-dimensional printing, also known as '3D printing', is a type of additive manufacturing technology that involves building layers of material on top of one another to build the final product. It functions similar to a printer, however instead of printing ink onto paper, it prints layers of material to construct a physical object.

The 3D printer scans the actual thing to be printed and then utilises the digital model as a blueprint to build the actual object. It builds microscopic layers by melting or softening a substance, such as plastic, and then extruding it through a nozzle. The item is progressively built-up layer by layer on top of the one before it until it is finished.

This process eliminates the need for expensive moulds or the assembly of multiple parts, making it a cost-effective choice. 3D printing also allows ideas to be prototyped and tested without undergoing an entire production process.



Applications of 3D Printing

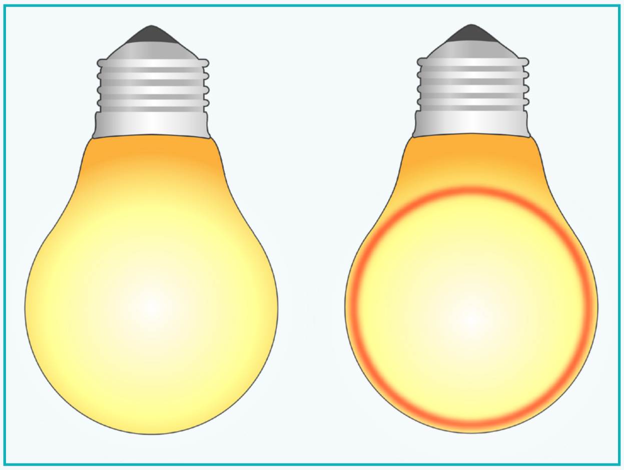


In Lesson 1, we started with modelling a Table Lamp. We created the base, arm, and the lampshade. In this lesson, we will continue with the activity from Lesson 1 and model the remaining parts. We will also learn some more 3D modelling concepts.

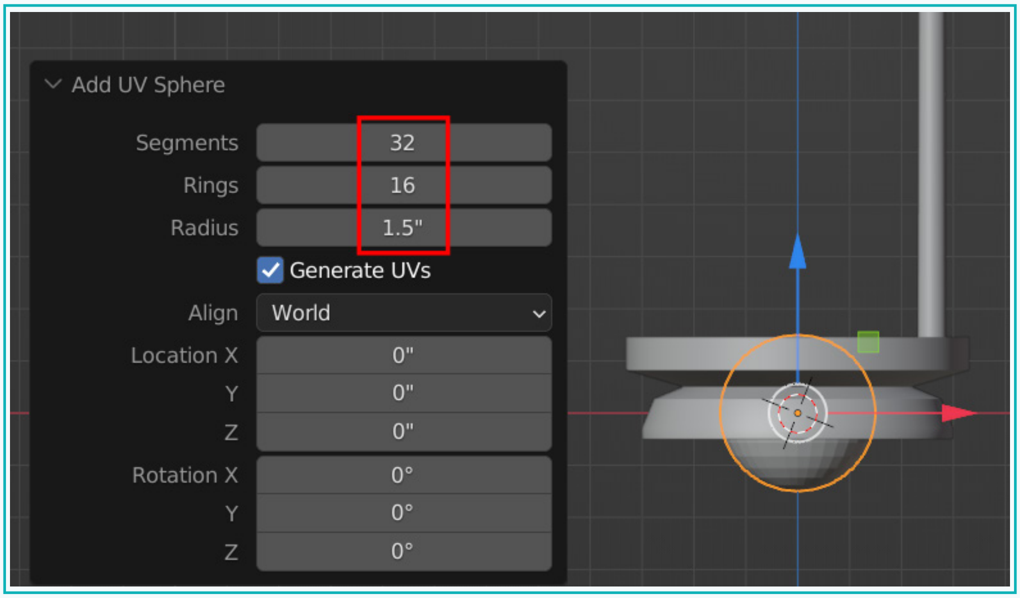
Activity

Create a Light Bulb for the Lamp

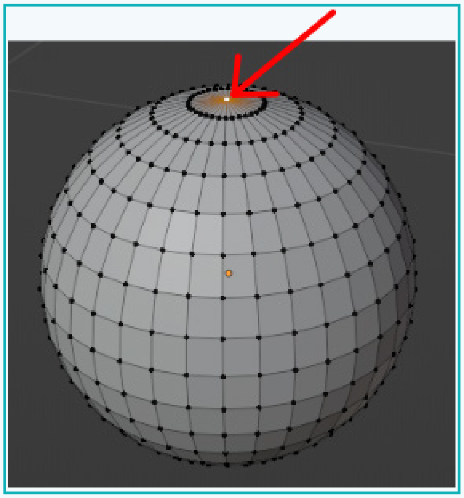
Let us create a light bulb that will fit within the lampshade. In the reference image below, we can see that the shape of the bulb is identical to a sphere. Hence, we will start modelling with the sphere.

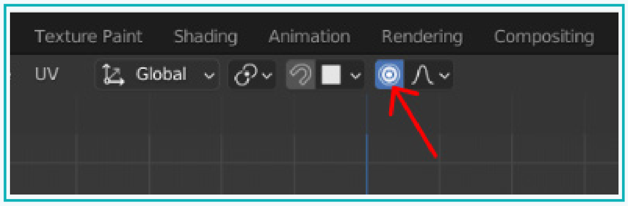


1. Create a 'UV Sphere'. Tab into the 'Edit' mode and select the 'Vertex Selection' mode. Click on the 'Proportional Editing' tool at the top.



Activity



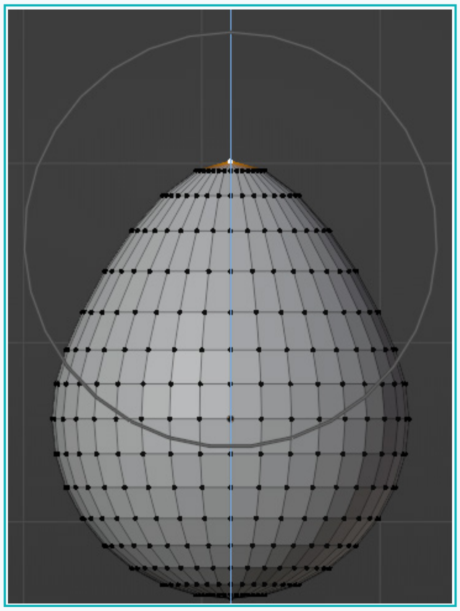


Proportional Edit

Proportional Edit is a technique that enables the transformation of the selected elements, such as vertices, while having an impact on adjacent elements. This means that altering the position of one vertex within a certain range can result in the movement of unselected vertices as well. This is particularly useful when we need to smoothly deform the surface of a mesh.

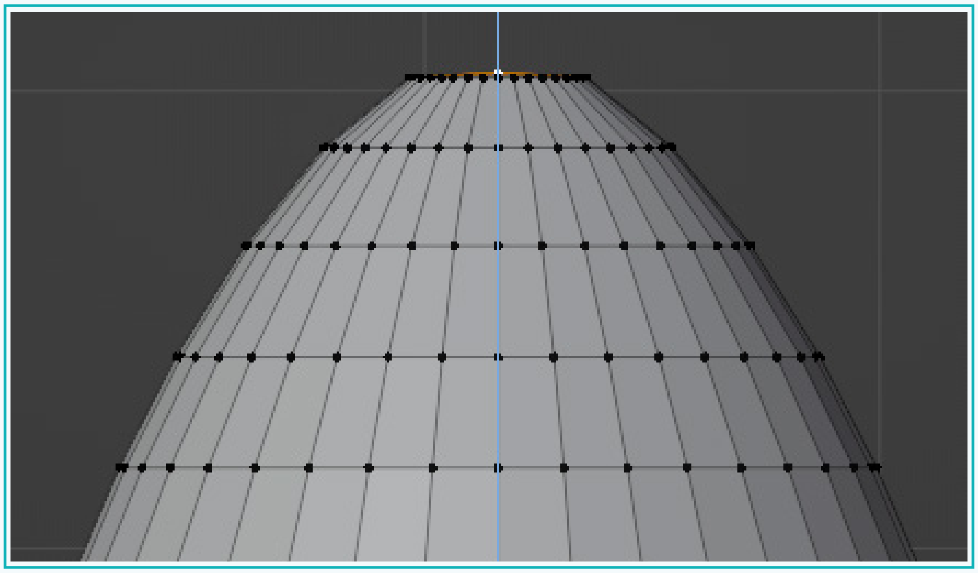
The radius of the proportional editing influence can be increased or decreased with the mouse wheel. As the radius is changed, the positions of the points surrounding the selection will get adjusted accordingly.

2. Press ‘G’ followed by ‘Z’ to move the selected vertex upwards proportionately. Scroll the mouse wheel to change the area of influence and get an oval shape.

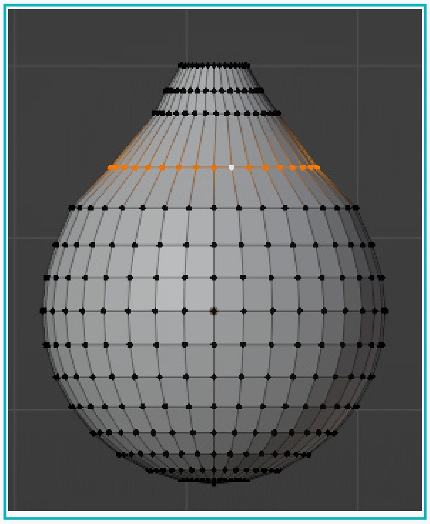


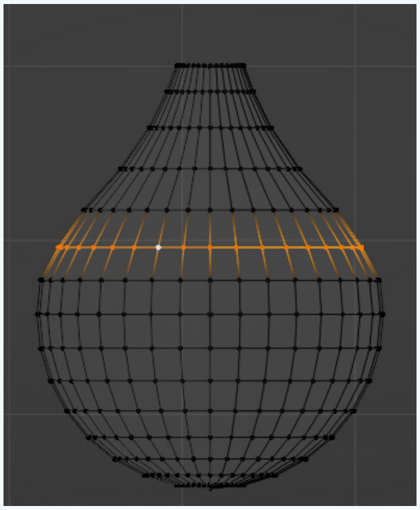
Activity

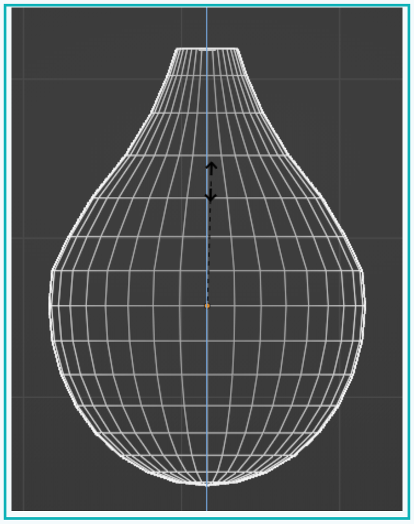
3 Switch off ‘Proportional Editing’ and move the selected vertex downwards, as shown.



4 Select edge loops and scale/move them to create the shape of a bulb.

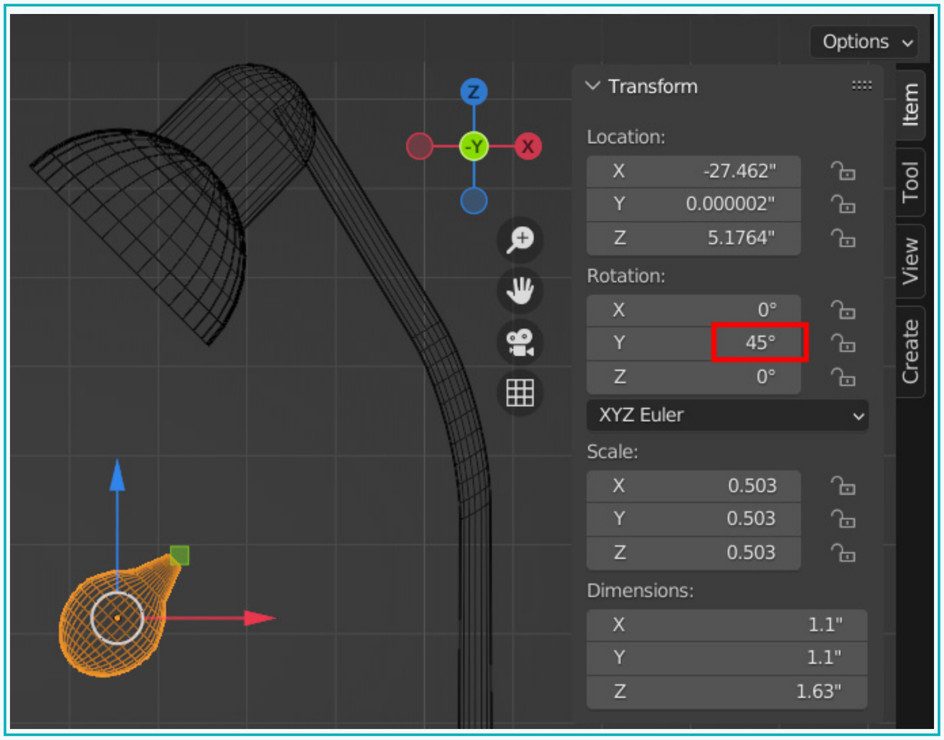


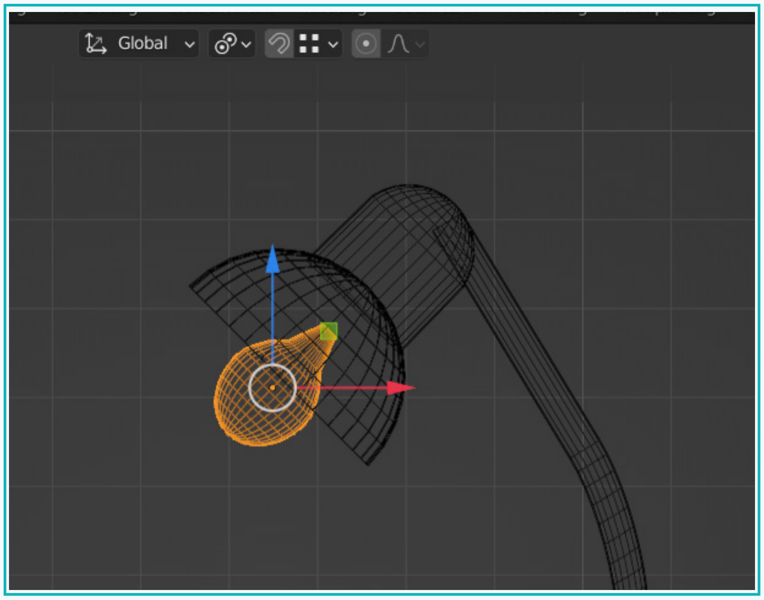




Activity

5 Change the 'Y-axis' rotation value to '45°' and position it into the Lampshade.

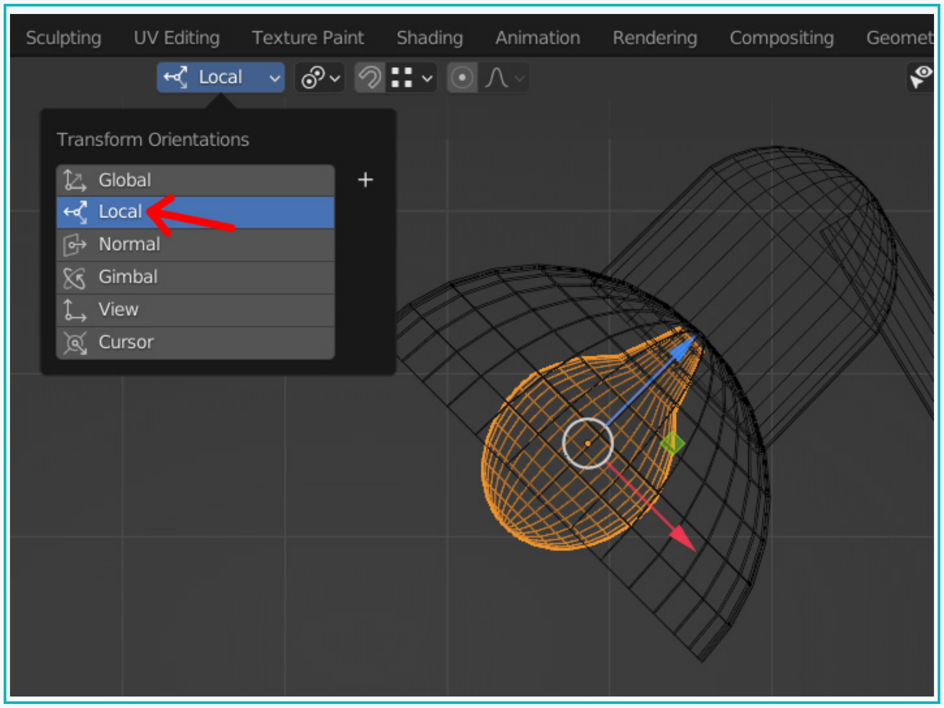




While positioning, try using the 'Local Tranform Orientation'.

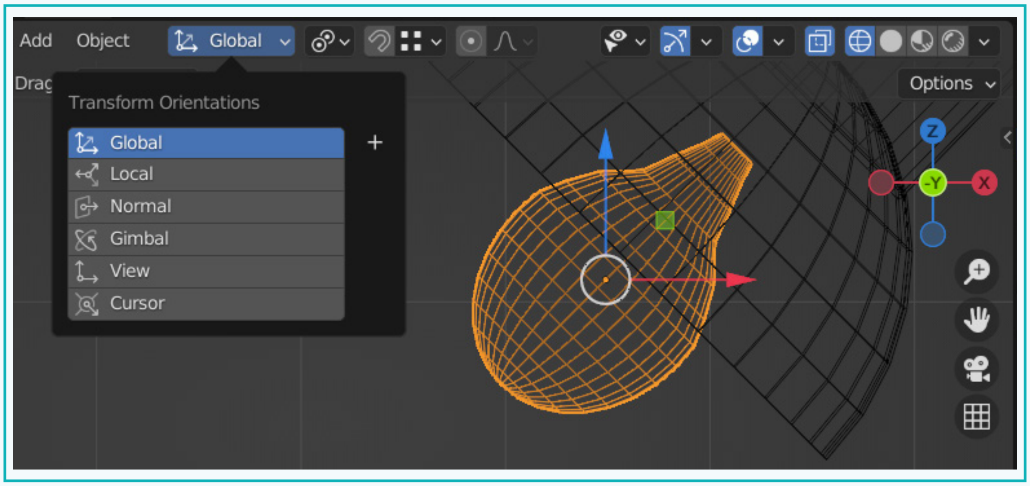
Activity

6 Click on the 'Transform Orientation' tool on the top, select 'Local' from the list.



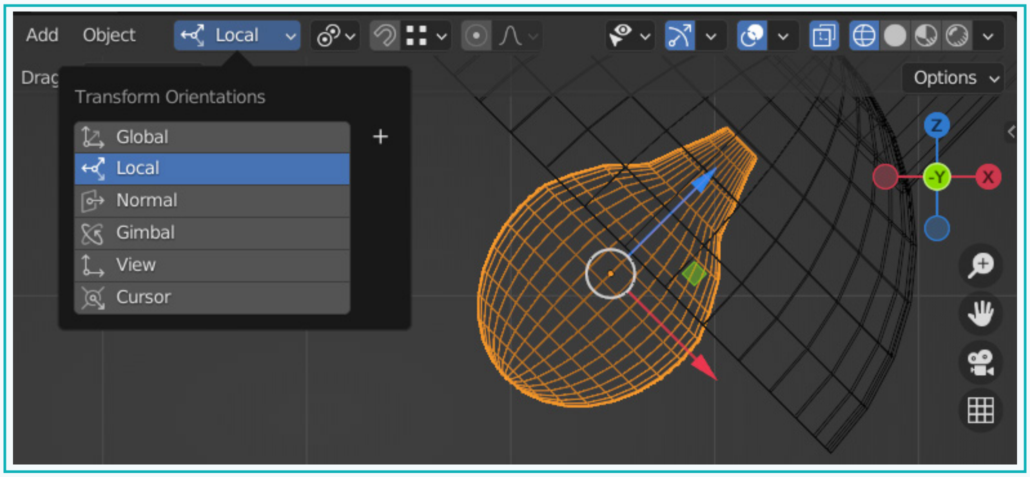
Notice the difference between the two options.

In 'Global Orientation', the Object Gizmo is always aligned with the global axes.



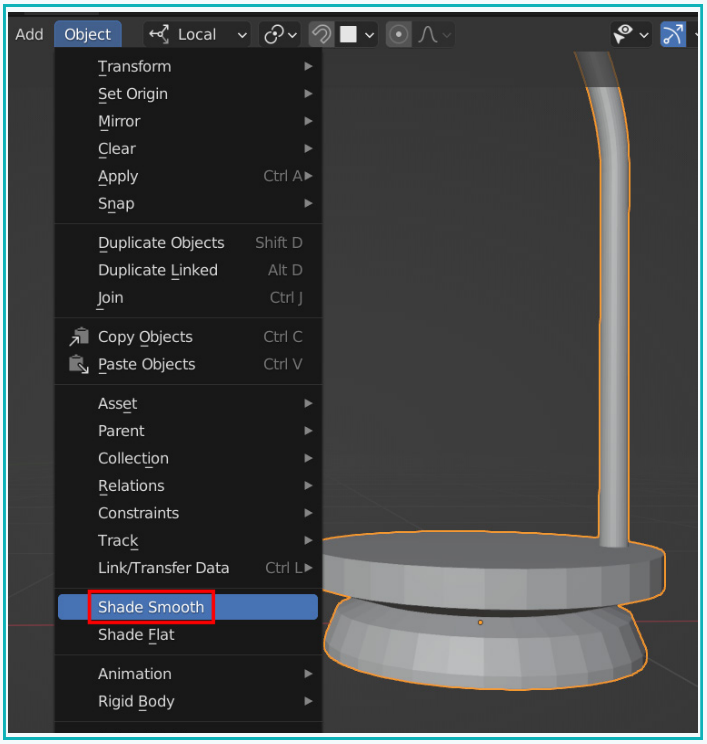
Activity

In 'Local Orientation', the Object Gizmo is always aligned with the local rotation of the object. Since the bulb has been rotated at 45°, the 'Move' handles correspond to the object's local rotation orientation.



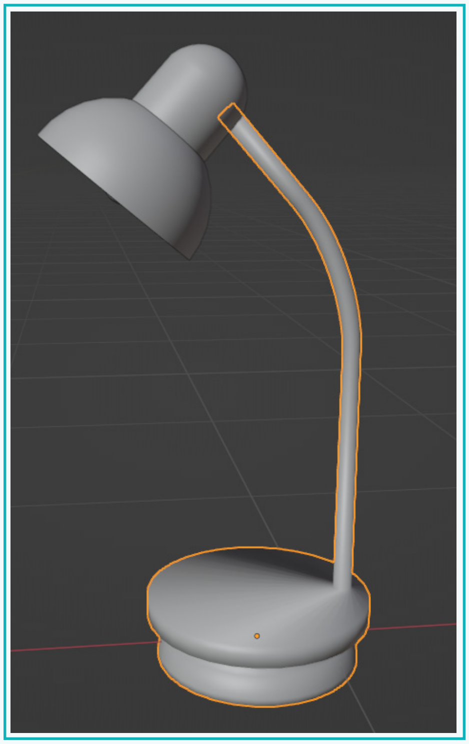
'Transform Orientation' is significant because it influences how the Move, Scale, and Rotate tools work. Basically, it defines the orientation of the Object Gizmo and makes it easier to make transformations in the directions you want.

7. If you notice the base, the faces on the side are visible. We can smooth them out by clicking on 'Object' → 'Shade Smooth'.

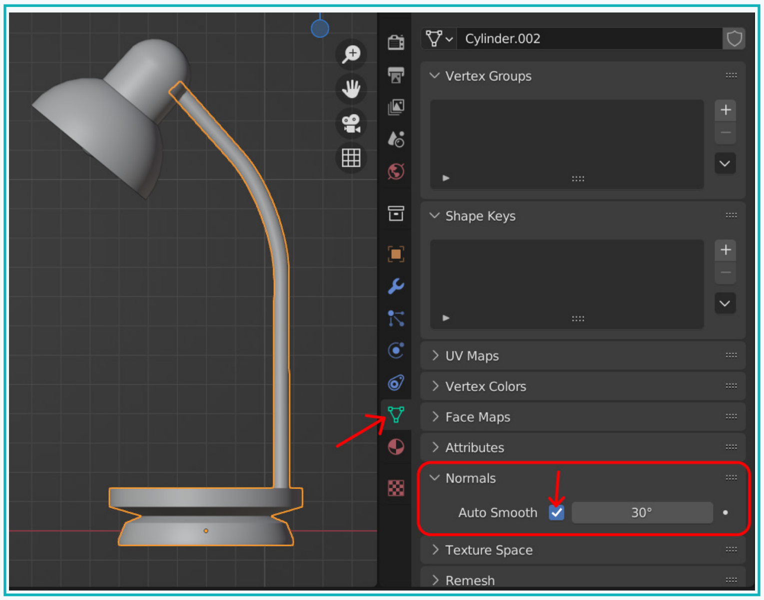


Activity

Sometimes while applying 'Smooth Shading', the direction in which a particular face is facing, gets inverted, giving the kind of effect, as seen in the image. This can be rectified by clicking on the 'Object Data Properties' category in the 'Properties' editor.



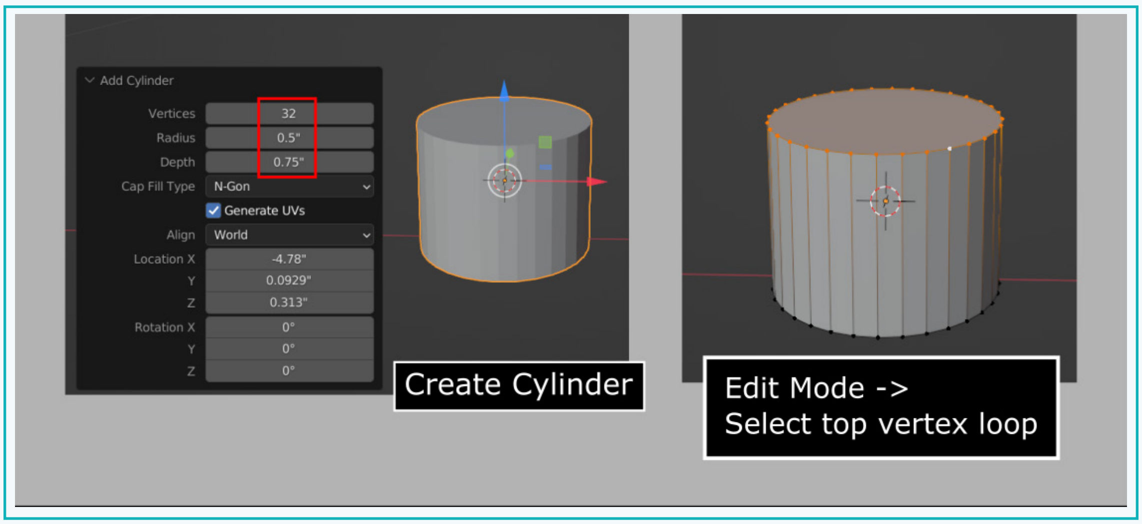
8 Click on the 'Object Data Properties' tab (inverted green triangle) in the 'Properties' editor. Under 'Normals', tick the 'Auto Smooth' option.

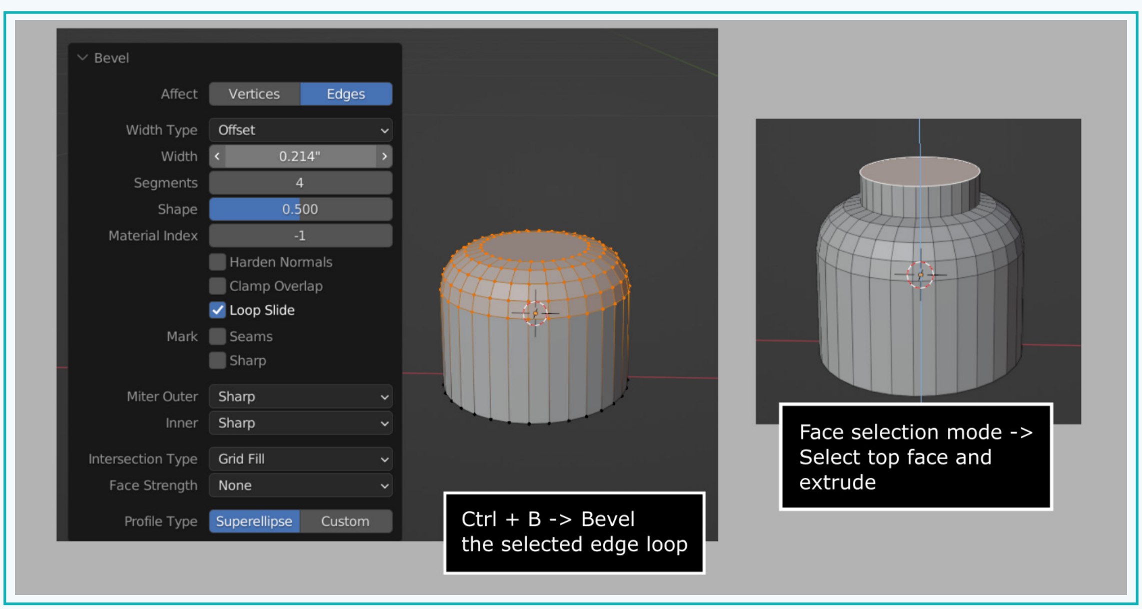


Activity

Create a Switch

Next, create a button at the base using a cylinder and extrude/bevel operations and position it at the base of the lamp.





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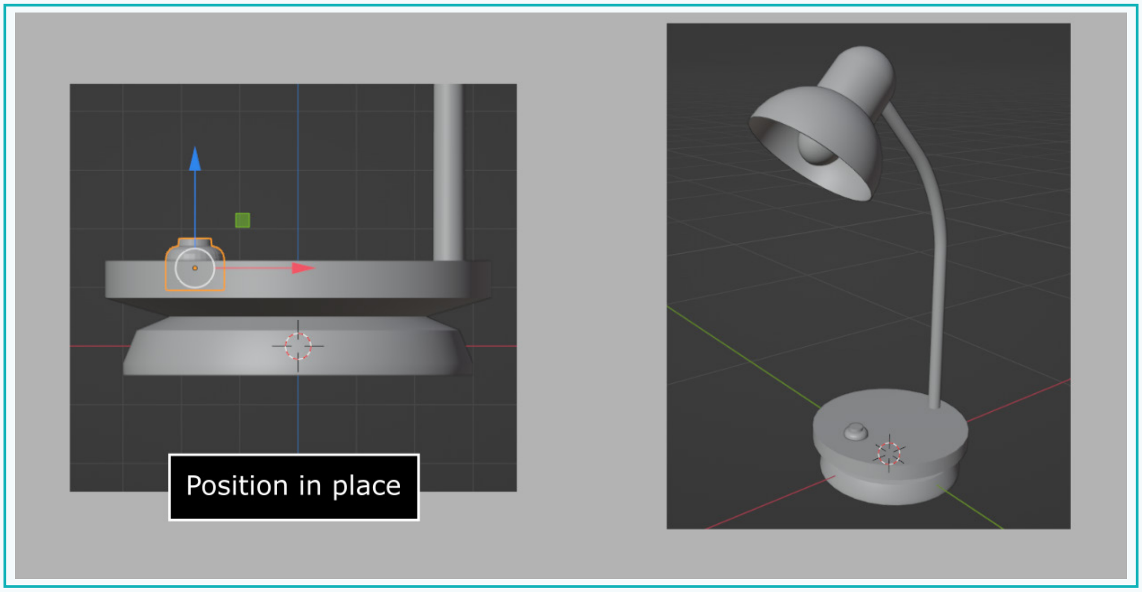


Exercise

State if the statements are True or False

1. ‘Proportional Edit’ does not have an impact on adjacent elements.

▲ ____________________

2. In ‘Local Orientation’, the Object Gizmo is always aligned with the global orientation of the object.

▲ ____________________

3. ‘Transform Orientation’ is important since it influences how the Move, Scale, & Rotate tools work.

▲ ____________________

Exercise

Fill in the Blanks.

4 The radius of the 'Proportional Editing' influence can be increased or decreased using the ____________________.

5 'Transform Orientation' defines the orientation of the ____________________ Gizmo.

6 In 'Global Orientation', the Object Gizmo is always aligned with the ____________________ axes.

7 The surface of an object can be smoothened out by applying ____________________ .

Answer the questions with one sentence.

8 What is the difference between Global & Local Orientation Transform?

▲ ____

____

9 What is 'Proportional Edit'?

▲ ____

____

Tick mark ☑ the Correct Answer

10 Which key can be used to select several objects?

a. Ctrl ☐ **b.** Shift ☐

c. Space ☐ **d.** Alt ☐

11 Which technique is useful when we want to smoothly deform the surface of a mesh?

a. Shade smooth ☐ **b.** Shade Flat ☐

c. Proportional Edit ☐ **d.** Solidify ☐

Activity

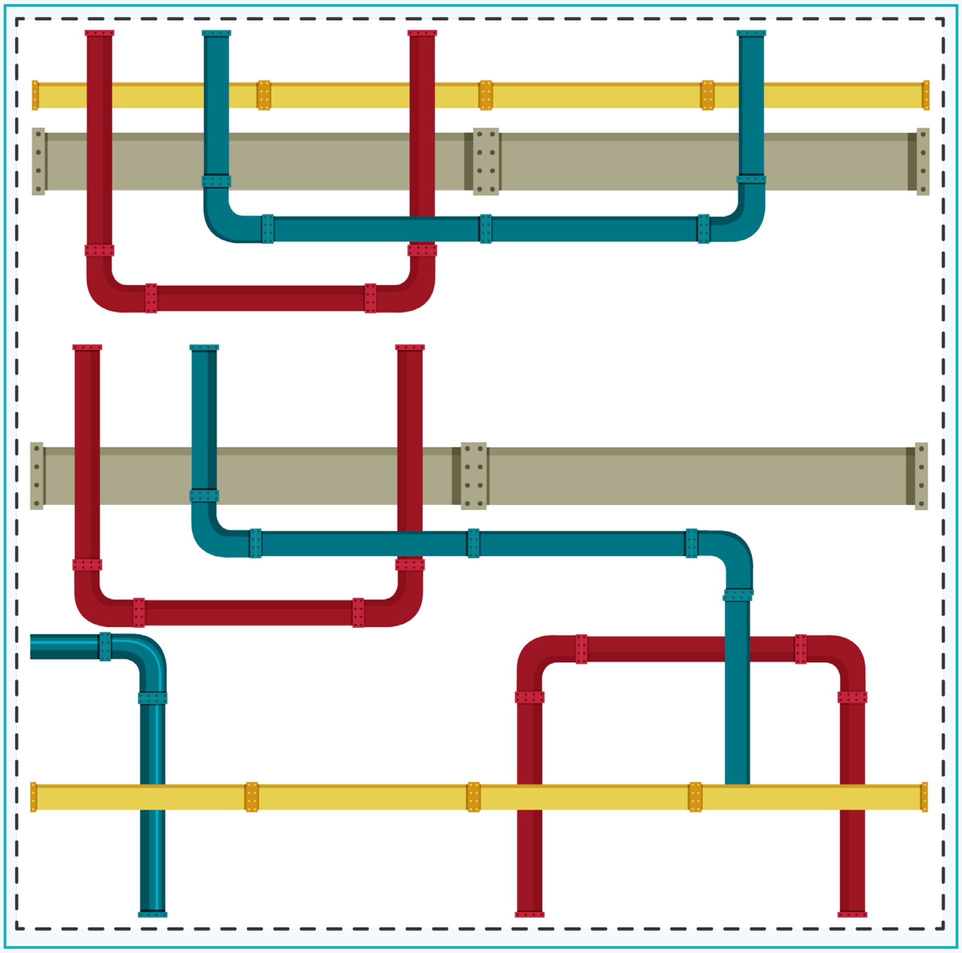
Create a Wire and a Plug

To create the wire of the lamp, we will use the ‘Curves’ tool.

Curve Tool

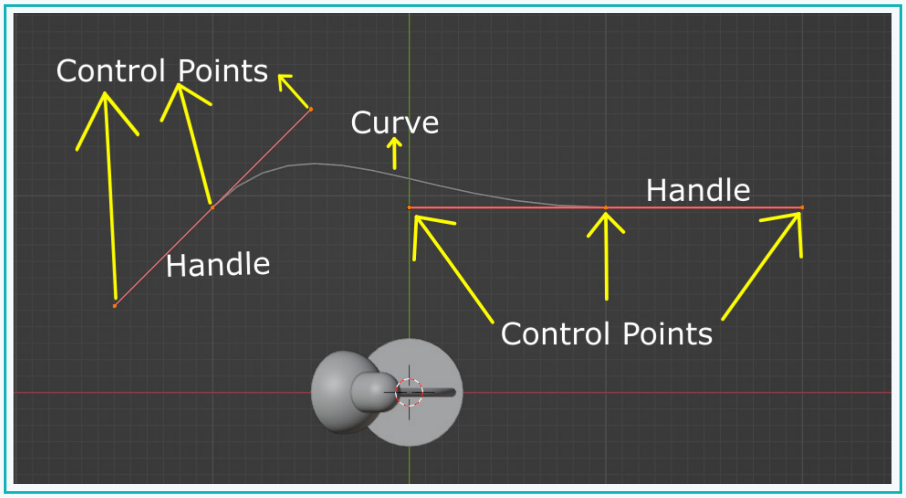
Blender’s ‘Curve’ tool lets the users create precise 3D shapes, providing us more flexibility and accuracy when creating complex models. By utilising curves instead of traditional geometric tools, we can attain a higher level of accuracy in our designs.

There are many instances where curves can be used for creating curved surfaces. Some real-world examples where the ‘Curve’ tool is used are for creating pipes and wires. Other applications of curves involve animating along a path using a curve.



Activity

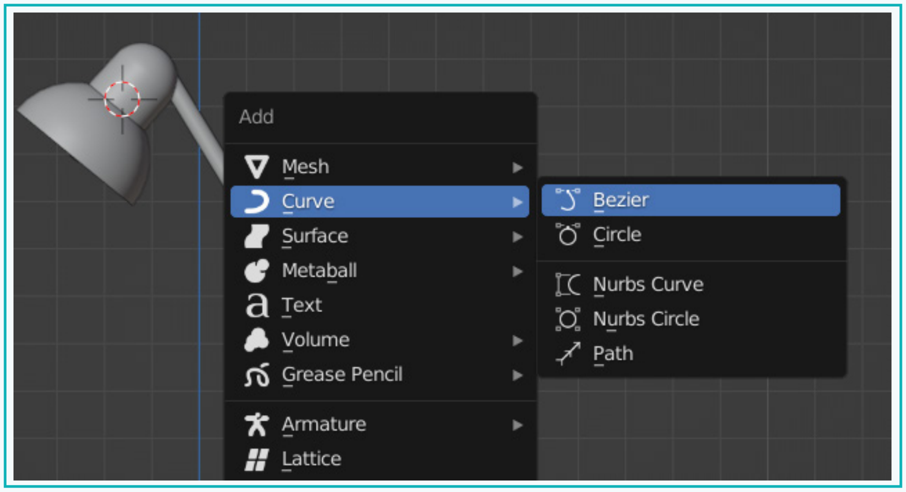
Curves have handles and control points to manipulate the curve object. The curves go through the control points to create the final result. Along with the control points, you can also move the handles to manipulate the shape of the curve.



Curve objects provide more control in some cases than mesh objects, as there is no need to alter or modify the mesh directly. Rather, by manipulating the control points on a curve, the mesh will get modified accordingly.

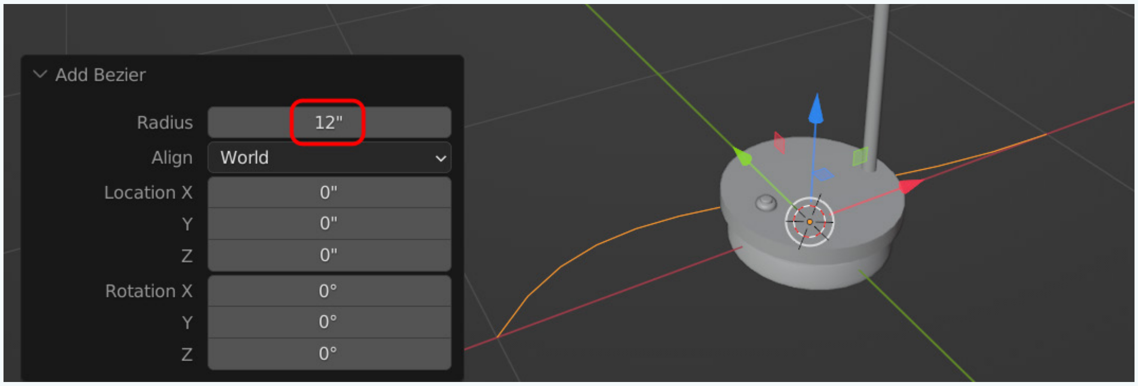
To create a curve object, click 'Shift + A' and go to the 'Curves' menu. There are 3 main types of curves - 'Bezier', 'Nurbs', and 'Path'. The 'Bezier' curve is the simplest. Let us create a 'Bezier' curve.

1. 'Press Shift + A' → 'Curve' → 'Bezier'.

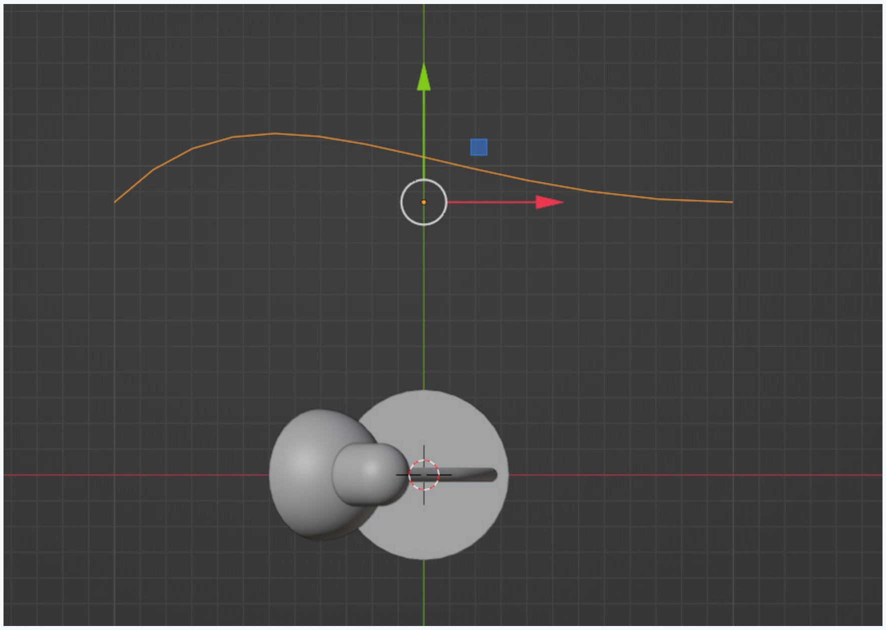


Activity

Create the curve with the following dimensions.



2 Go into the 'Top' view (**'Numpad 7'** or **'Backtick (`)'** → **'Top'**) and move it where it can be viewed clearly.



The curve object can be modified or transformed in 'Object' mode as well as in 'Edit' mode. Through object mode, one can move, scale, and rotate the entire curve using the standard tools. When we are in the 'Object' mode, we can also adjust extrusion, bevelling, and other properties in the 'Object Data Properties' tab in the 'Properties' window.

Activity

Under 'Geometry', one can adjust the bevel of the curve object. There are three settings for bevelling the curve object:

a. Round: It turns the curve into a round tube-like object. The resolution and smoothness of the bevel can be adjusted. The end of the curve can be sealed by clicking the 'Fill Caps' button.

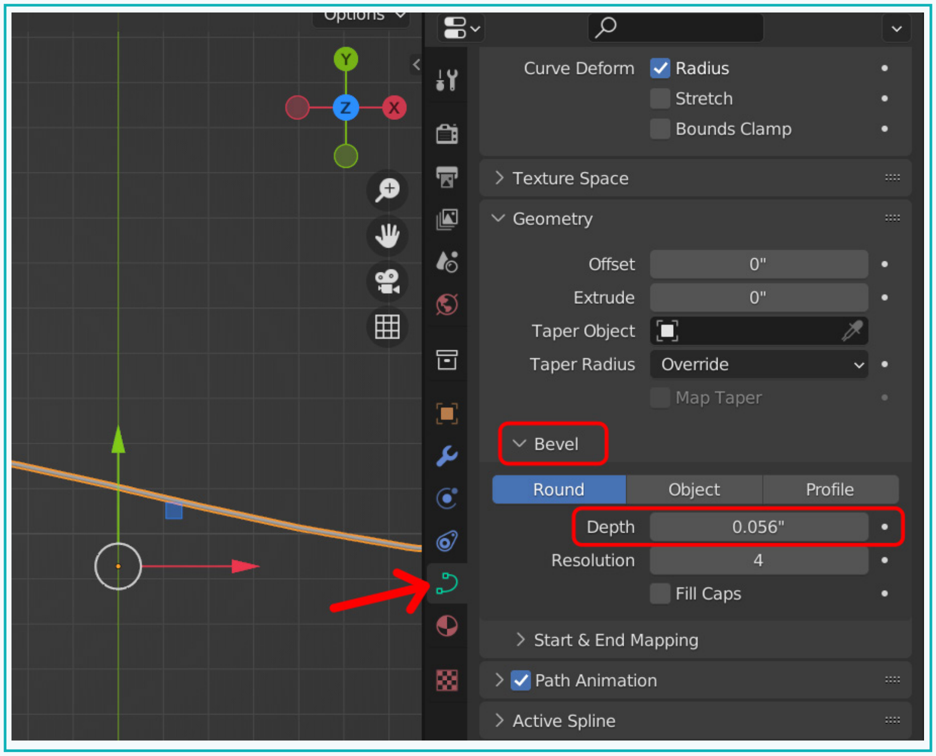
b. Object: It allows one to extrude the curve based on another curve object selected.

c. Profile: This similarly transforms the curve into a tube-like object, but it allows one to define the profile of the bevel by adjusting the line on the 'Widget' window. This could be helpful when creating more intricate curve details.

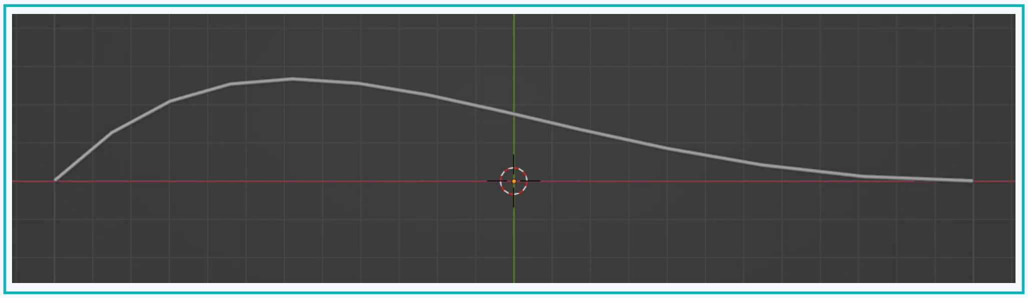
To turn the curve into a round tube-shaped object, let us modify the 'Depth' property under the 'Round' settings, as shown below.

3 Click on the 'Object Data Properties' tab in the 'Properties' window.

- Under 'Bevel' → 'Round', increase the 'Depth' property value. We can see the curve turn into a pipe/wire shape. Set the 'Depth' value to a desired value.



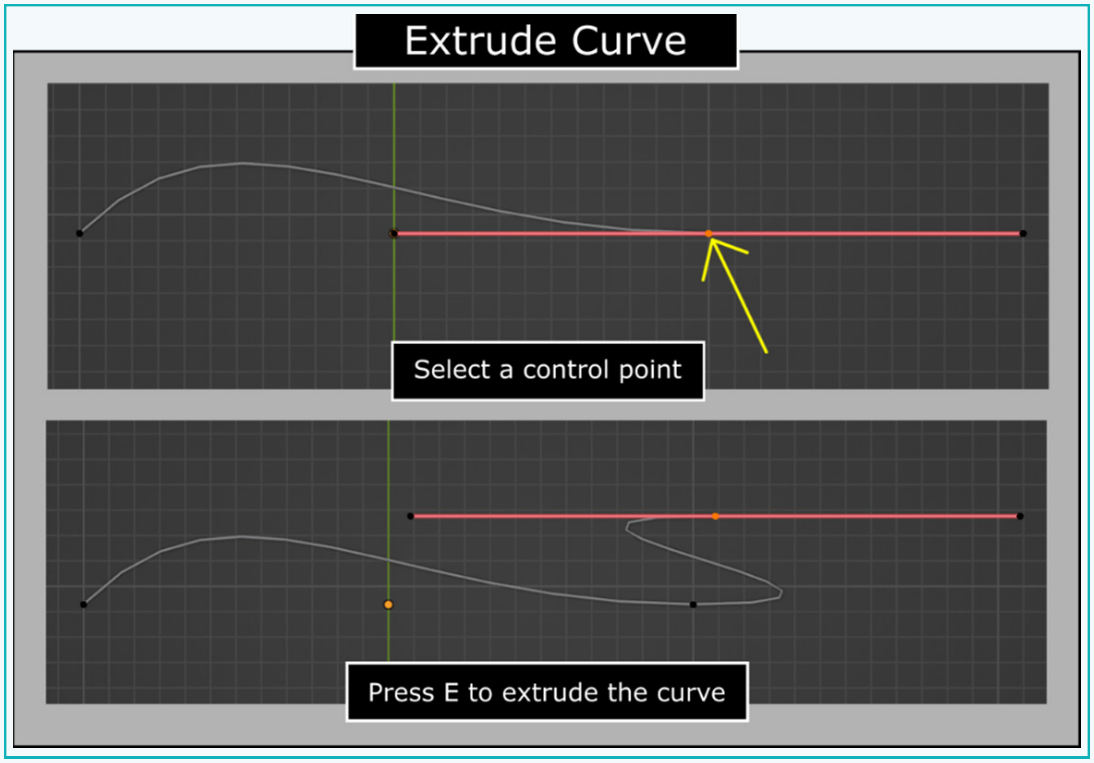
Activity



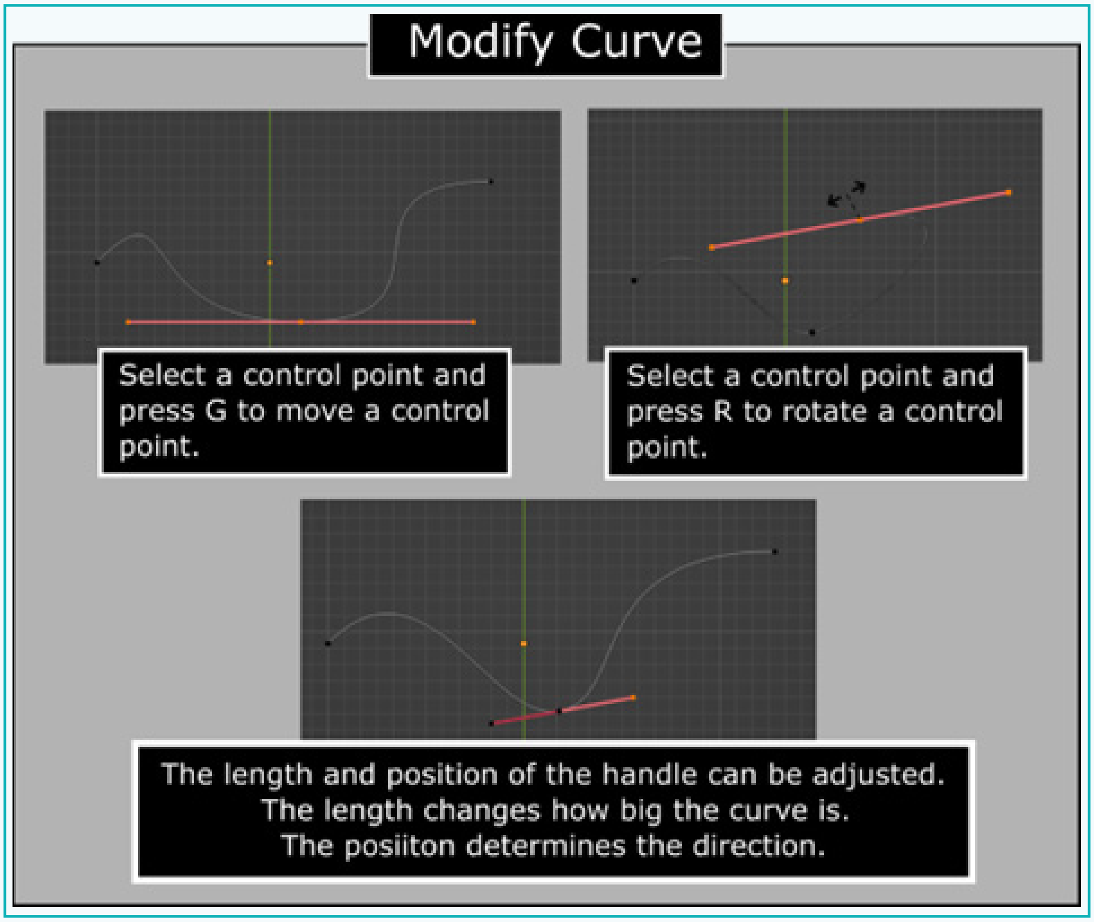
To modify the shape of the curve, we must adjust the control points and handles of the curve. For this, we need to switch to the 'Edit' mode.

4 Press 'Tab' to go into the 'Edit' mode.

The regular Select and Transform tools can be used on the control points to deform the curve. The new control points can be extruded by pressing 'E' to extrude.



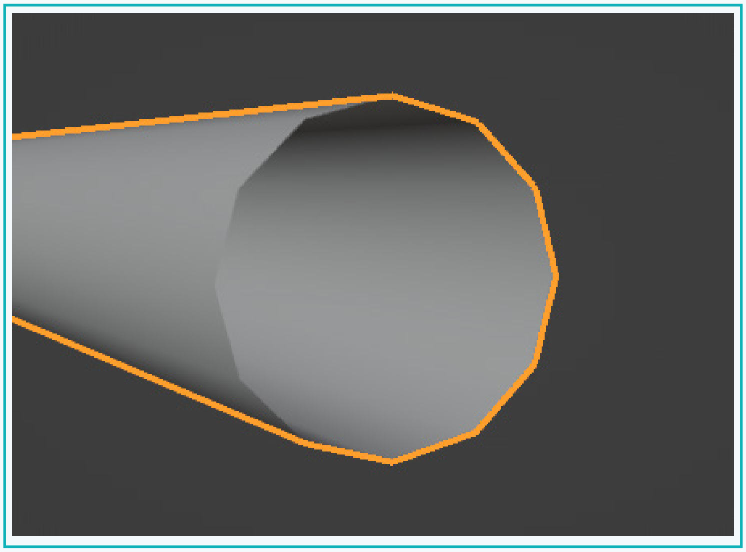
Activity



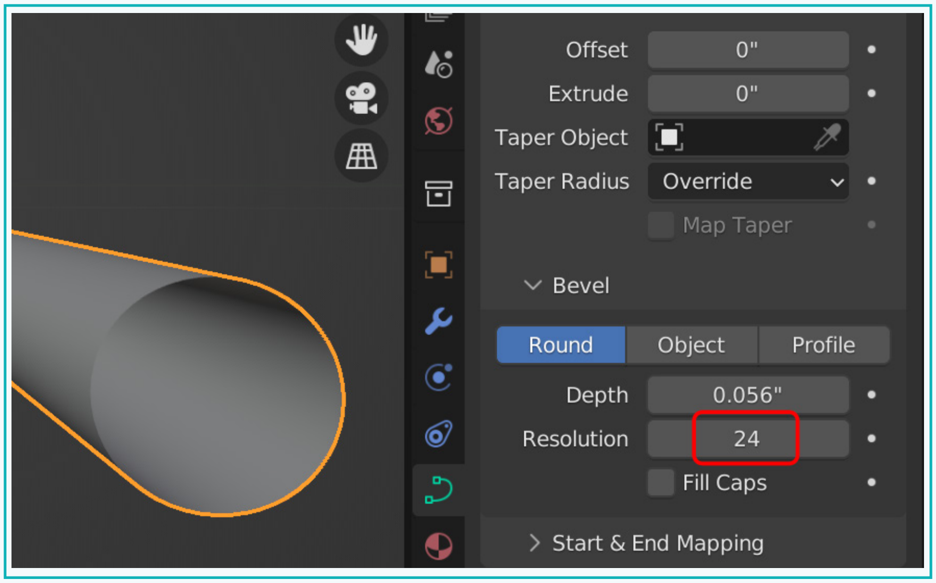
- Get an optimal shape as desired for the wire of the table lamp.

Activity

5 Go into the ‘Object’ mode. Zoom into the edge of the curve. You will notice that the edge is not smooth.

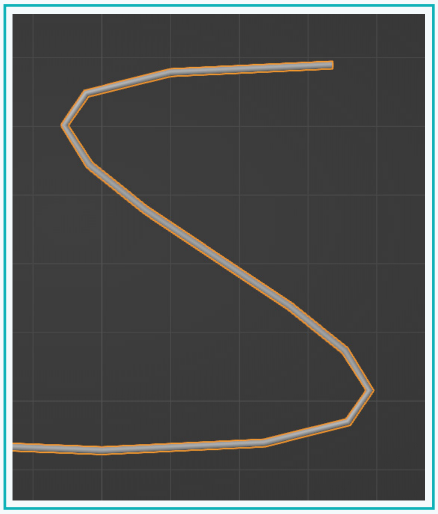


6 In order to make it smooth, increase the ‘Resolution’ property under the ‘Bevel’ section. As the value increases, the edge starts getting smoother.

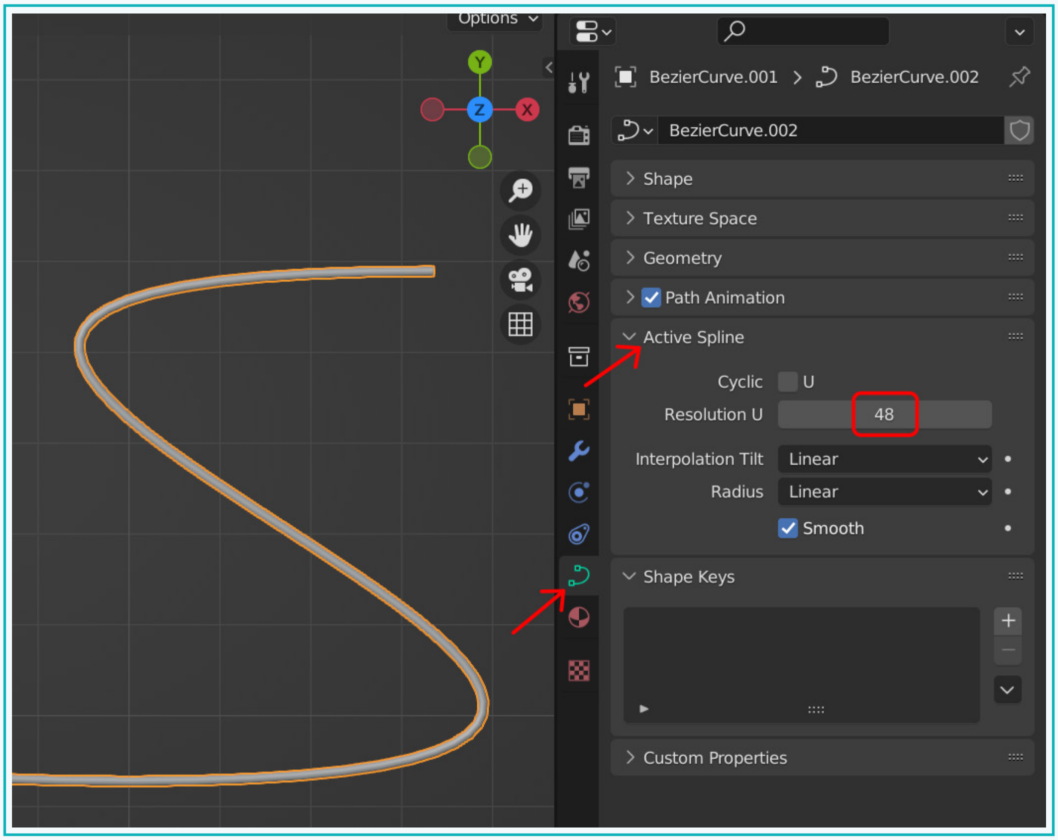


Activity

Also, if you look at the curve, the surface is uneven and not rounded, as curves should be.



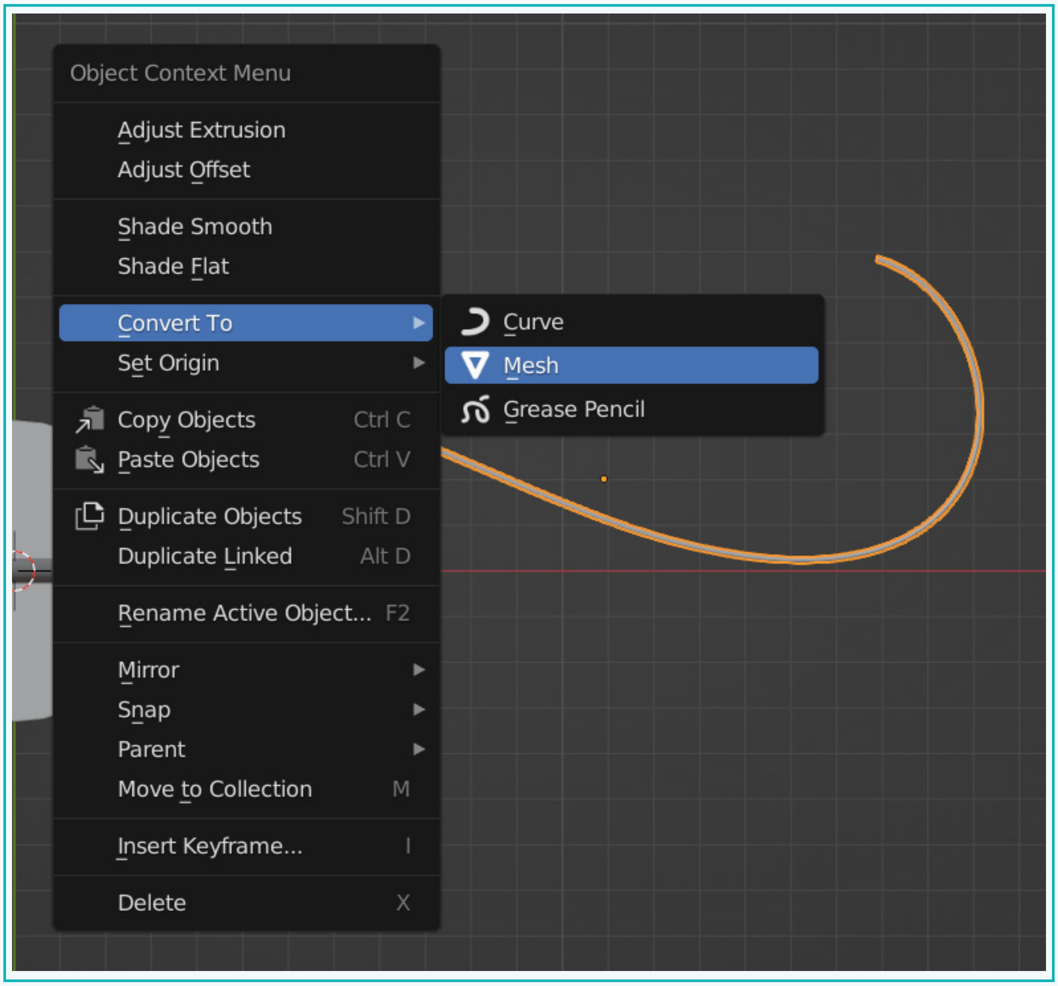
7 For this, under ‘Active Spline’, increase the ‘Resolution U’ property. You will notice the curve starts getting smoother.



Activity

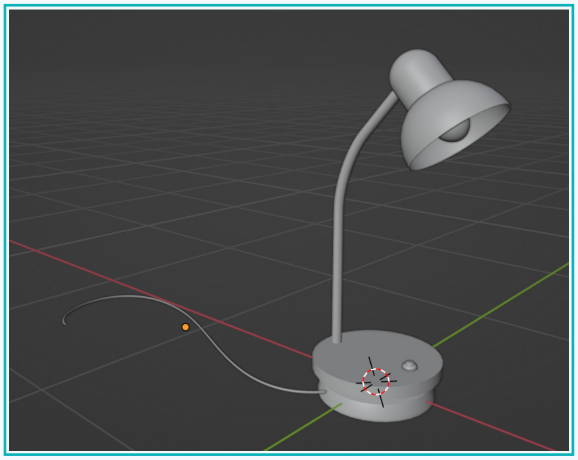
Finally, once all the changes to the curve have been made, we can transform the curve into a mesh.

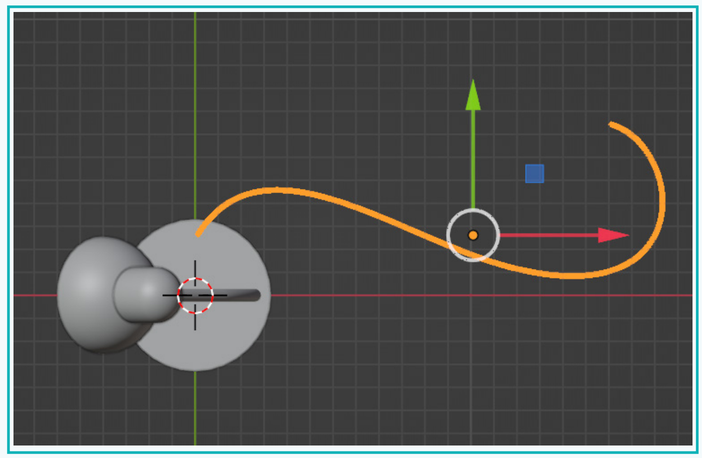
8. Tab into the ‘Object’ mode, right-click and select ‘Convert To’ -> ‘Mesh’. This will convert the curve into a mesh. This applies any settings made to the curve object in the ‘Object Data’ property tab.



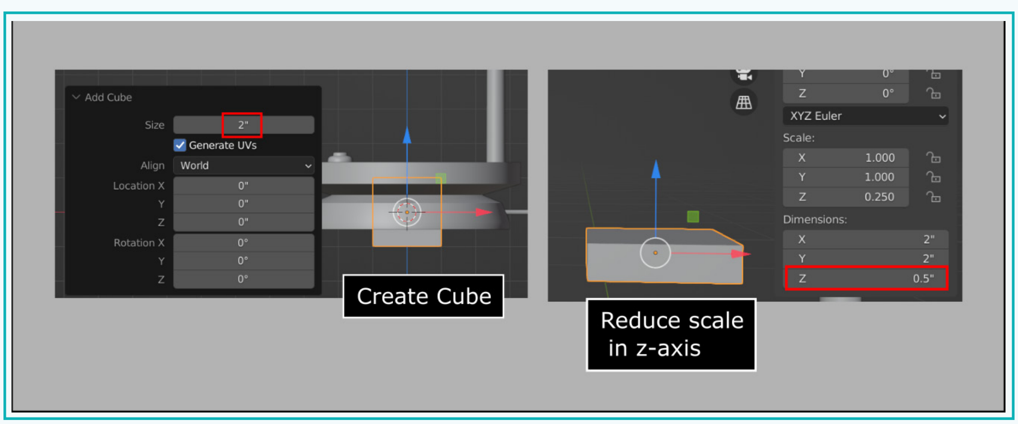
Activity

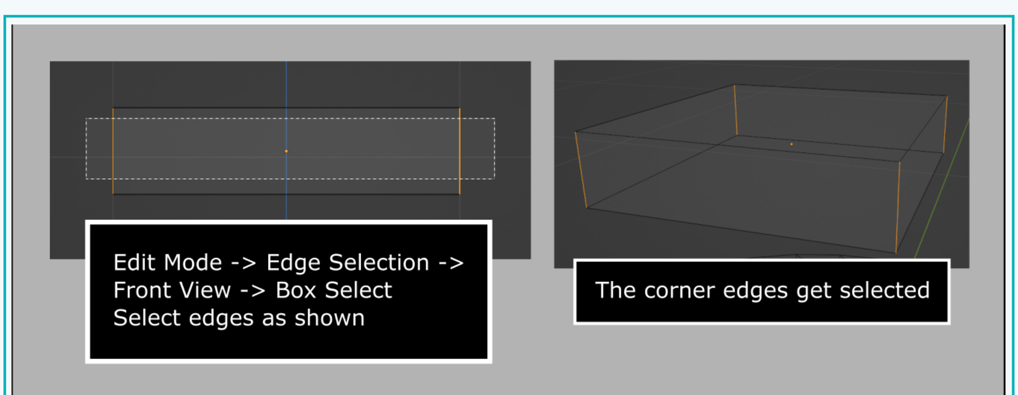
9 Position it such that it is attached to the base of the lamp.



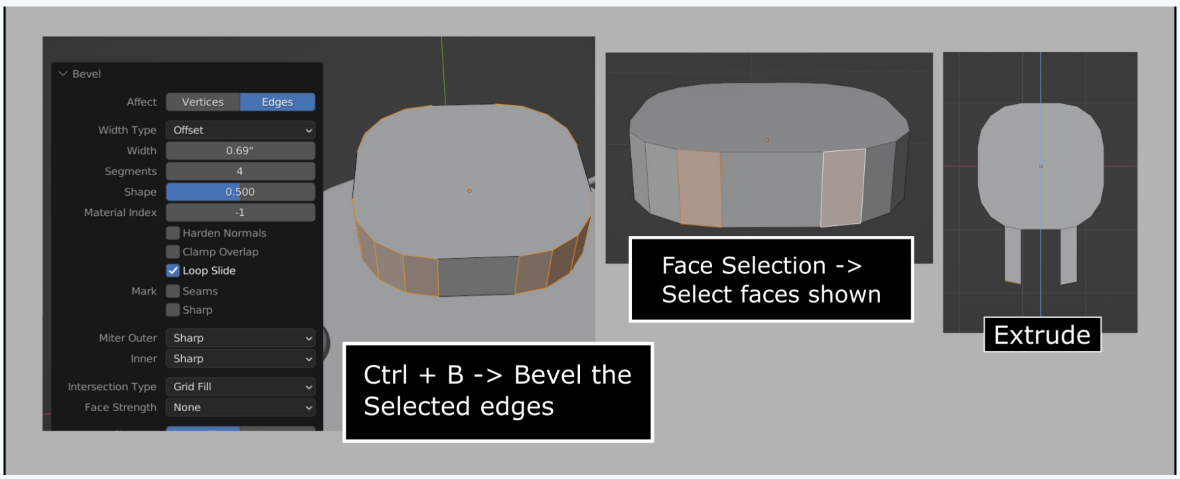


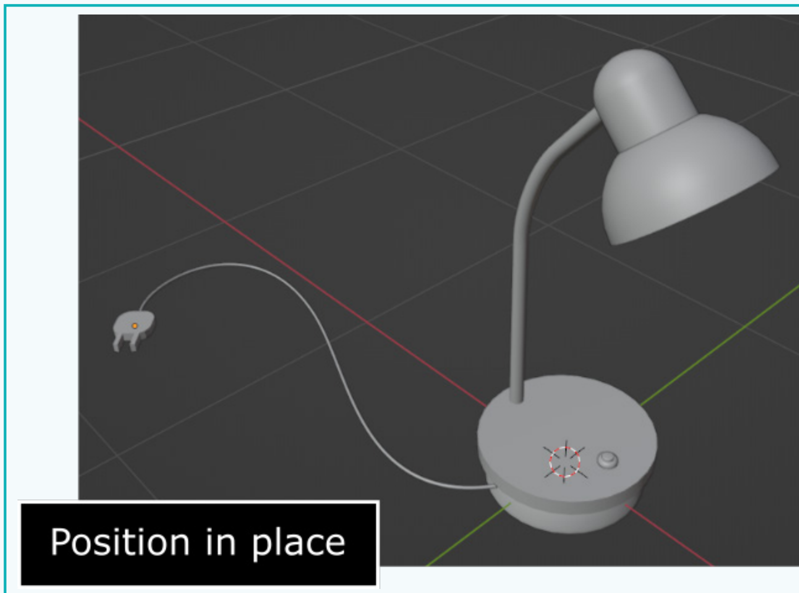
Next, create a plug using a cube and some basic operations that we have learnt so far.





Activity


Bevel
Affect Vertices Edges
Width Type Offset
Width 0.69"
Segments 4
Shape 0.500
Material Index -1
Harden Normals
Clamp Overlap
Loop Slide
Mark Seams
Sharp
Miter Outer Sharp
Inner Sharp
Intersection Type Grid Fill
Face Strength None
Face Selection -> Select faces shown
Extrude
Ctrl + B -> Bevel the Selected edges

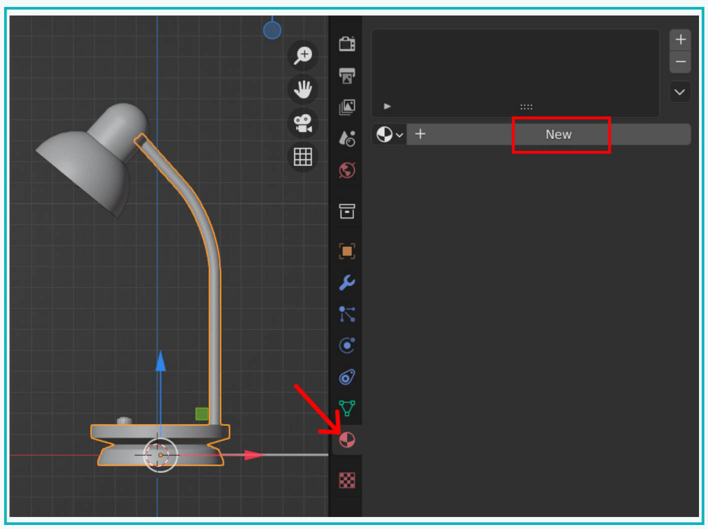

Position in place

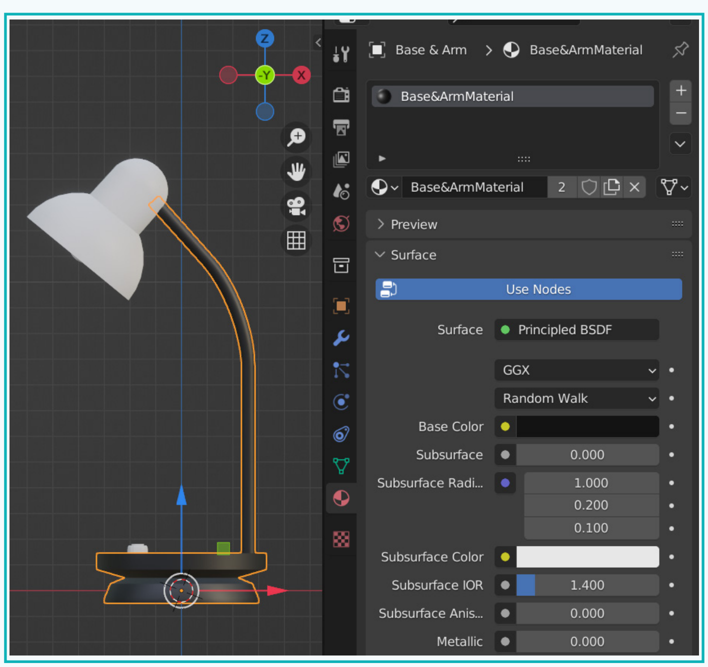
Activity

Add Materials

Next, let us make it look attractive by adding materials to it.

1. Select the base and neck of the lamp. Click on the ‘Material Properties’ category in the ‘Properties Editor’ and click on ‘New’. Change the base colour. Make sure to be in the ‘Material Preview’ mode to be able to see the materials applied.

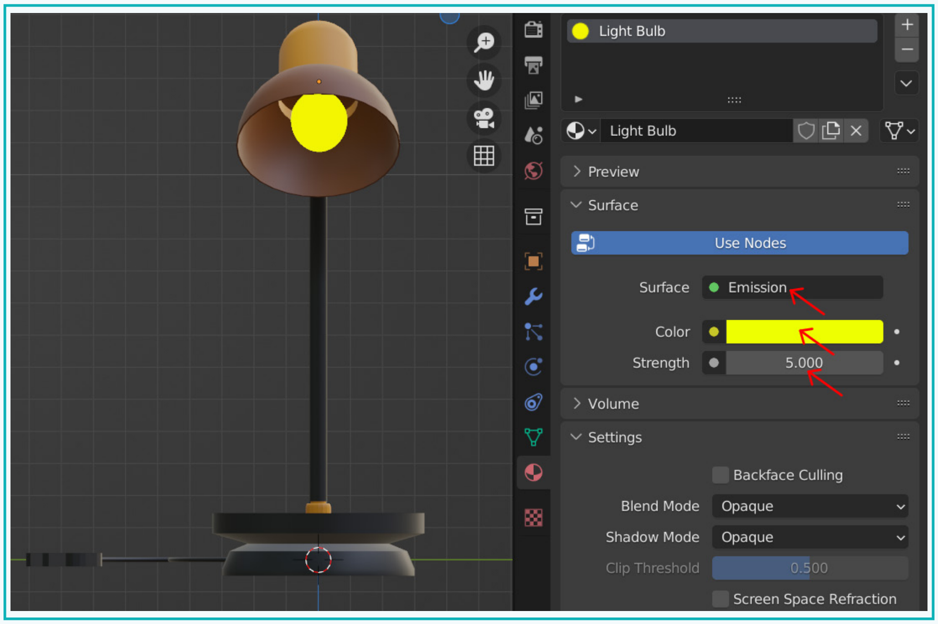




Activity

2. Similarly, add materials to the other parts of the Lampshade, the button on the base, as well as the wire.

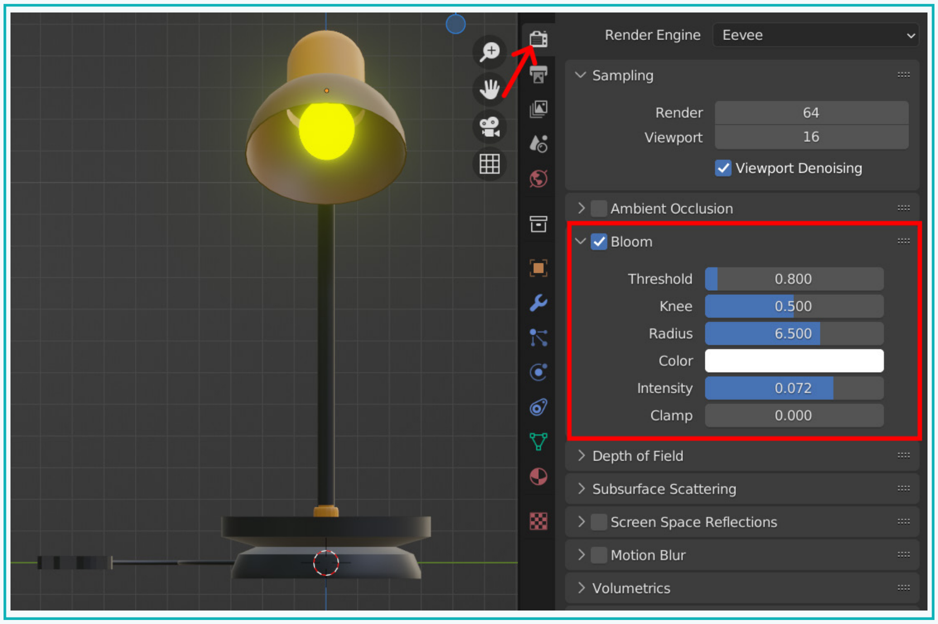
3. For the bulb, select it and add a material of the type 'Emission'. An object with 'Emission' as the shader, lights up objects around it. Change the colour and 'Strength' values.

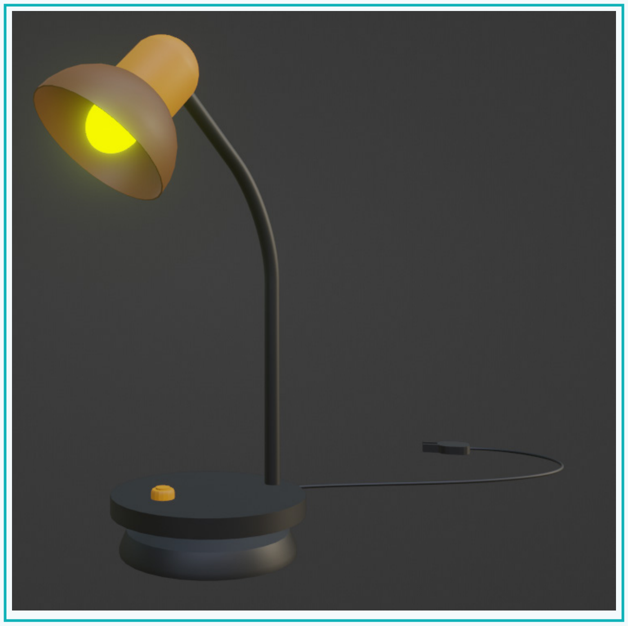


Activity

4 To give a glow effect to the bulb, go into the 'Render Properties' category and select the 'Bloom' property. The 'Bloom' property gives a perception of very bright light.

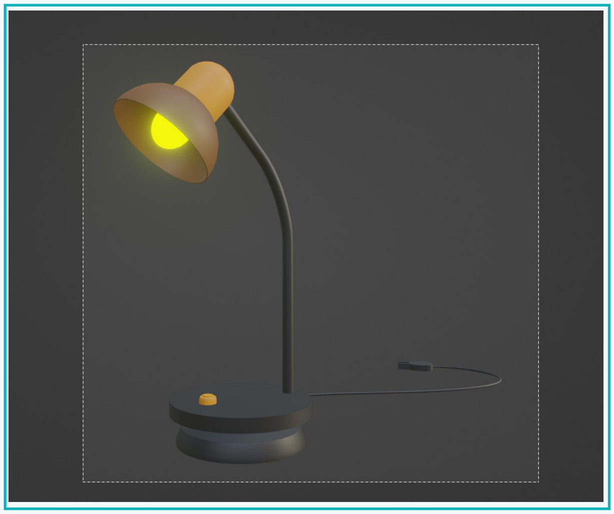
Adjust the values under it and experiment with the glow effect.

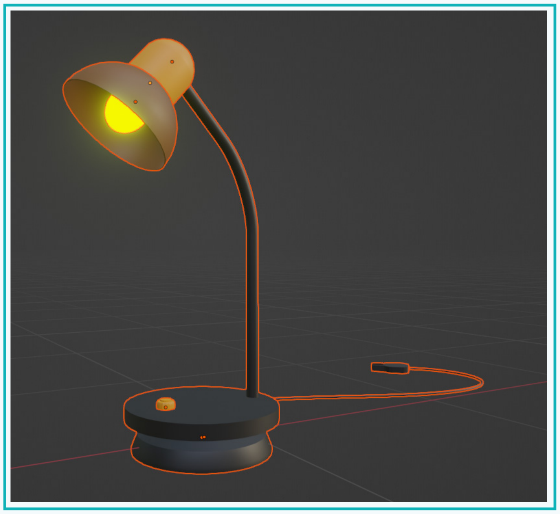


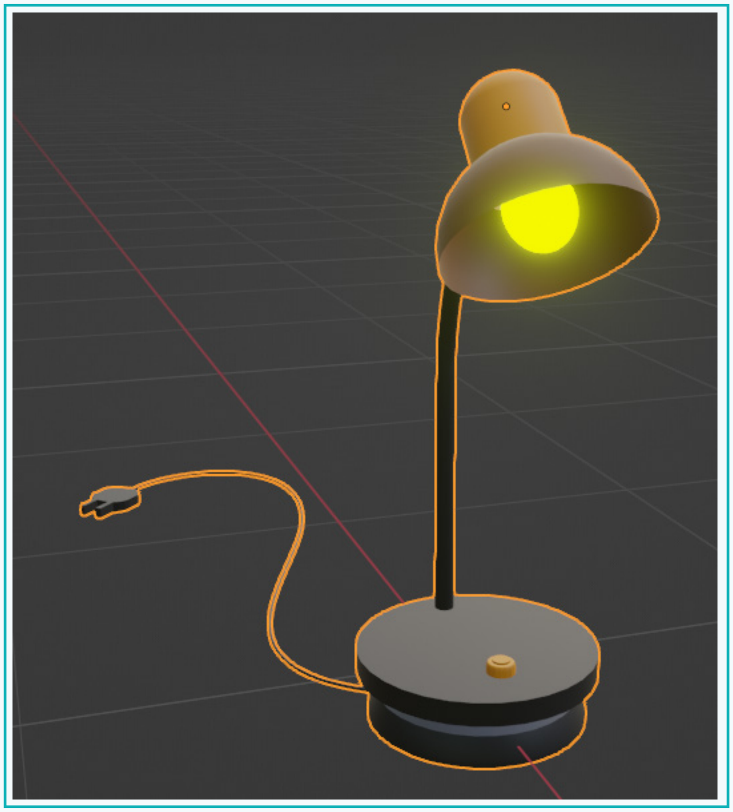


Activity

5 Using ‘Box Select’, select all the parts of the table lamp. Press ‘Ctrl + J’ to join all the parts together to form one mesh.







Exercise

State if the statements are True or False

12 The 'Solid Display' mode helps us to preview applied materials.

▲ ______________________

13 Curve objects can be modified only in the 'Edit' mode.

▲ ______________________

14 It is possible to convert a curve into a mesh.

▲ ______________________

15 Curves can be extruded.

▲ ______________________

Fill in the Blanks.

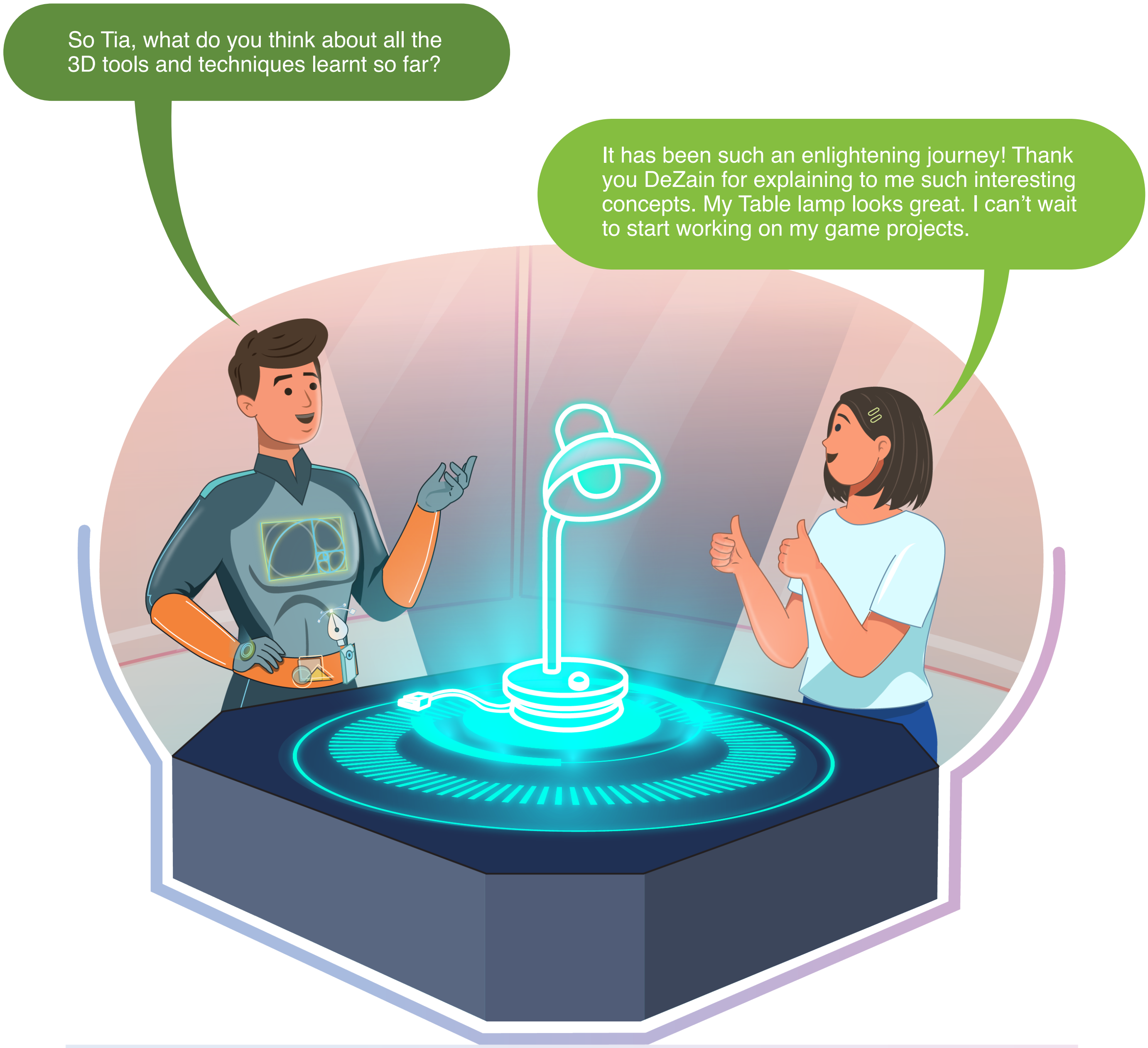
16 The __________________ shader on an object illuminates the objects surrounding it.

17 The __________________ property helps in giving a glowing effect to objects.

18 Curves have __________________ and __________________ to manipulate the curve object.

19 The _____ of the handle determines how big the curve is.

In today's session, we learnt about 3D printing and its applications. We discussed new concepts such as proportional edit, transform orientation, and curve tool. We completed building the Table lamp model successfully.



Let's continue to learn and grow in our 3D design journey. Who knows, maybe one day we'll see our creations in popular games.

Tech Flash

- **Proportional Edit** : It is a technique that allows you to manipulate specific elements, such as vertices, while having an impact on adjacent elements.
- **Transform Orientation** : It defines the orientation of the Object Gizmo and makes it easier to make transformations in the directions you want.
- **Curve tool** : It is useful in creating complex models like pipes, tubes, wires, etc. as well as in animating along a path.
- **Emission Shader** : It illuminates the objects around it.
- **Bloom property** : The 'Bloom' property gives a perception of very bright light.